

Project 2 — Example: Mean age of California high schools

STAT 109 illustrative sample — use your own population and variable for your project

Sample report for Project 2: Estimation from a random sample

This page is a worked example for the structure of [Project 2](#). Your submission must use your own population and variable.

Download: [Sample one-page write-up with histogram \(PDF\)](#)

Sample notebooks (CA high schools):

- **Mean with FPC:** [project-2-estimation-mean-fpc.ipynb](#) — [Open in Colab](#)
 - **Proportion with FPC:** [project-2-proportion-fpc.ipynb](#) — [Open in Colab](#)
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Introduction

What is the mean age of high schools in California? As someone who teaches college-level classes, I was interested in learning more about the population of schools my students might come from. One question is how old those schools tend to be: I think older schools might tend to have more resources and stronger connections with a community, whereas newer schools might have less resources.

Parameter: mean age (in years, in 2024) of all California high schools.

Population: all high schools listed in the California School Directory.

Methods

Data collection

On April 3, 2024, I visited www.cde.ca.gov/SchoolDirectory and searched for any type of high school. I found $N = 1617$ schools.

I used the [random number generator on RossmanChance.com](#) to choose $n = 10$ numbers from 1 to 1617. I located each corresponding school by setting the number of schools shown on a page to 100, going to the page with the right hundreds value, and counting up or down to find the school matching each random number's hundreds and ones digits.

For each of the 10 schools, I looked up the Open Date and calculated age in 2024 by subtracting the open year from 2024. My variable is numerical (age in years).

Data analysis

I used the `survey` package in R to compute a 95% confidence t -interval with finite population correction for the mean age of all CA high schools, using the ages from my sample of 10 schools. Step-by-step code is in the [mean notebook](#).

Conditions for this interval ([checklist in Read 2](#)):

1. SRS from a numbered list of the population.
2. Mean / CLT: population ages roughly normal or $n > 30$. Because $n = 10$, I rely on the ages of all schools in California being roughly normal).

3. FPC: finite $N = 1617$, SRS without replacement $\rightarrow$ multiply SE by $\sqrt{(N-n)/(N-1)}$.

Results

Descriptive statistics

The mean age of the ten schools was 35 years and the standard deviation was 25 years. See Figure 1 in the [sample PDF](#) (histogram of ages).

Inferential statistics

Based on my dataset, I am 95% confident that the mean age of all CA high schools is between 20 and 51 years old in 2024.

Conclusion

The mean age of high schools in CA was younger than I expected. Because my sample of 10 schools was chosen with a simple random sampling scheme, I do not expect my estimate to be biased. However, $n = 10$ is small, so the estimate is not very precise: there is about a 31-year spread between the lower and upper bounds of the interval. I could improve precision by increasing the sample size.

Related: proportion example

The [proportion notebook](#) uses the same school ages but estimates the proportion of CA high schools that are more than 50 years old in 2024, a second Project 2-style analysis but with a binary outcome and FPC.